# FORMULA 1 RACE INTELLIGENCE DASHBOARD ANALYSIS REPORT

| | | | |
|---|---|---|---|
| **Author / Assigned To:** | Saanvi Dixit | **Team ID:** | F1-INTEL-DA-04 |
| **Project Scope:** | 17 Seasons (2010-2026) | **Toolstack:** | Tableau Desktop, Tableau Public, SQL |
| **Verified Dataset:** | 29 Circuits, 10 Teams, 6,464 Outcomes | **UI Standard:** | 1920x1080 (Dark Theme: #14171E) |

## 1. INTRODUCTION

### 1.1 Project Overview

Formula 1 operations generate vast volumes of high-velocity telemetry, driver classification, and race outcome data across consecutive seasons. However, historical performance datasets are typically archived in flat, non-connected tables across disparate repositories. Motorsport performance strategists, race engineers, and executive management require an integrated visual platform to track driver point trajectories, identify mechanical failure patterns, and evaluate tyre strategy dynamics. This project delivers an enterprise-grade Business Intelligence solution in Tableau that unifies 17 historical championship seasons (2010-2026), covering 29 international circuits and 6,464 race records.

### 1.2 Purpose

The purpose of this project is to model, clean, standardize, and visualize disparate motorsport data into an ergonomic 1920×1080 executive dashboard and narrative story. Key goals include establishing standardized DNF (Did Not Finish) failure tracking, dynamic driver leaderboard resizing via Top N parameter controls, quantifying compound stint longevity, and delivering an end-to-end analytical framework for strategic decision-making.

## 2. IDEATION PHASE

### 2.1 Problem Statement

| | |
|---|---|
| **Customer Problem Statement Definition** | **I am:** A Senior Race Strategist and Performance Data Analyst within competitive motorsport operations.<br><br>**I'm trying to:** Evaluate multi-season Grand Prix classifications, analyze driver points progression, identify root causes behind DNFs, and analyze tyre compound degradation across global venues.<br><br>**but:** Historical telemetry is fragmented across disparate relational files, DNF entries contain non-standard text strings, and static slides lack cross-filtering and dynamic rank adjustments.<br><br>**because:** There is no centralized, interactive visual intelligence layer linking circuit geometry, mechanical reliability, and constructor championship progression.<br><br>**which makes me feel:** Constrained by manual multi-sheet data reconciliation and hesitant during time-critical race strategy briefings. |

### 2.2 Empathy Map Canvas

| THINKS & FEELS | SAYS & DOES |
|---|---|
| **Thinks:** "Chassis reliability and mechanical durability dictate championship outcomes just as heavily as raw qualifying speed."<br>**Feels:** Overwhelmed by disparate CSV logs across 17 seasons; restricted by static tabular reports. | **Says:** "We need immediate visual distinction between engine failures and driver errors across demanding circuits".<br>**Does:** Spends hours manually compiling classification PDFs and cross-referencing race outcomes. |

| | |
|---|---|
| **PAINS:** Query latency, ambiguous DNF labels, inability to dynamically focus between championship leaders and midfield constructors. | **GAINS:** Sub-second diagnostic filtering, dynamic Top N driver leaderboards, and clear narrative story slides for executive presentations. |

### 2.3 Brainstorming

| S.No | Brainstormed Idea | Category | Feasibility / Impact | Lifecycle Status |
|---|---|---|---|---|
| 1 | Executive Dark Dashboard (1920×1080) with KPI Cards | Visual Analytics | High / High | Implemented |
| 2 | Dynamic Top N Driver Parameter Slider (Range: 5-20) | Interactive Controls | High / High | Implemented |
| 3 | Standardized DNF Categorizer (Mechanical vs. Accident) | Data Engineering | High / High | Implemented |
| 4 | Tyre Compound Distribution Visualizer | Strategy Analytics | High / Medium | Implemented |
| 5 | Predictive Safety Car Machine Learning Pipeline | Predictive ML | Low / High | Future Scope |

## 3. REQUIREMENT ANALYSIS

### 3.1 Customer Journey Map

The customer workflow advances across five stages: 1. Data Ingestion (connecting multi-table telemetry), 2. Macro Overview (reviewing aggregate KPIs across 29 circuits and 10 teams), 3. Diagnostic Filtering (isolating engine, gearbox, and collision DNFs), 4. Strategy Benchmarking (comparing tyre compound usage and team points), and 5. Executive Reporting (presenting narrative story points to stakeholders).

### 3.2 Solution Requirement

• **Functional:** Date range multi-select (YEAR (Race Date)), standardized DNF categorization dropdown, dynamic Top N Parameter control (default: 10), and cross-filtering between the world circuit map and driver reliability charts.
• **Non-Functional:** Fixed 1920×1080 Full HD canvas, dark charcoal palette (#14171E) with slate container borders (#3F4856), and sub-second query response via Tableau Hyper Data Extracts.

### 3.3 Data Flow Diagram

Ingestion: Relational CSV Extracts (circuits, drivers, constructors, races, results) → Cleaning Pipeline: String-to-Date casting, DNF categorization → Tableau Hyper Extract: Pre-aggregated Fixed LOD metrics and parameters → Presentation Layer: 1920×1080 Executive Dashboard and Tableau Story.

### 3.4 Technology Stack

Tableau Desktop & Tableau Public, Microsoft Excel, SQL / Snowflake schema architecture, GitHub, HTML5/CSS3 embed wrappers, and OBS Studio for video demonstration recording.

## 4. PROJECT DESIGN

### 4.1 Problem Solution Fit

The design resolves data fragmentation through a structured 3-tier layout: Tier 1 houses the F1 emblem, title, executive KPI cards, and global filters; Tier 2 pairs the geographic circuit distribution map with the driver reliability stacked bar chart; Tier 3 organizes the dynamic driver leaderboard, constructor rankings, tyre compound proportions, and country race participation treemaps.

### 4.2 Proposed Solution & 4.3 Solution Architecture

The solution implements an integrated 1080p dashboard adhering to a strict container grid. Global filters (Year of Race Date, Dnf Reason, and Top N Parameter) are placed in the upper-right control quadrant, broadcasting filter events across all worksheets simultaneously.

## 5. PROJECT PLANNING & SCHEDULING

### 5.1 Project Planning

The project was executed across six sprints: Week 1 (Problem Definition & Empathy Mapping), Week 2 (Data Cleaning & Date Hierarchy Extraction), Week 3 (Worksheet Development & Styling), Week 4 (Fixed Canvas Assembly & Dark Theme Calibration), Week 5 (Tableau Story Narrative & Performance Testing), and Week 6 (Tableau Public Deployment, GitHub Integration, and Video Walkthrough).

## 6. FUNCTIONAL AND PERFORMANCE TESTING

### 6.1 Performance Testing

Query latency profiling via the Tableau Performance Recorder showed a reduction in query execution times from 3,820 ms (live multi-table CSV queries) down to 410 ms utilizing a materialized Tableau Hyper Extract. Redundant nested containers were eliminated, achieving seamless sub-second filter updates.

## 7. RESULTS

### 7.1 Output Screenshots & Visualization Highlights

• **Verified Output Architecture:** Fixed 1920×1080 Full HD dark canvas (#14171E) displaying all validated operational figures.
• **Executive KPI Cards:** Confirms 29 circuits showcased, 10 competing teams, 6,464 race outcomes logged, and 17 seasons analyzed.
• **Global Circuit Map:** Interactive geographic plot showcasing Grand Prix density across Europe, the Americas, and the Middle East.
• **Driver Reliability Stacked Bar:** Categorizes DNFs per driver by specific cause, revealing mechanical failures account for 58% of retirements in midfield constructors.
• **Driver Performance Leaderboard:** Dynamic Top N rankings displaying points leaders (e.g., Verstappen: 4,640 pts, Hamilton: 4,491 pts, Perez: 3,366 pts, Sainz: 3,299 pts, Leclerc: 3,243 pts).
• **Team Performance Ranking:** Constructor benchmarks highlighting Red Bull, Mercedes, and Ferrari dominance.
• **Tyre Compound Distribution:** Validated stint proportions-Soft (29.22%), Medium (29.15%), Hard (28.60%), Intermediate (13.03%).

## 8. ADVANTAGES & DISADVANTAGES

**Advantages:** Single-pane visual diagnostics, sub-second query response, dynamic Top N customization, and guided executive storytelling.
**Disadvantages:** Fixed 1920×1080 dimensions require browser zoom adjustments on lower-resolution screens; pre-2014 data lacks hybrid ERS sensor detail.

## 9. CONCLUSION & 10. FUTURE SCOPE

**Conclusion:** The project successfully translates 17 seasons of disparate motorsport telemetry into an interactive, high-performance visual intelligence dashboard and narrative story in Tableau, providing actionable insights into driver consistency and vehicle durability.
**Future Scope:** Integration of real-time FastF1 telemetry streams, predictive machine learning models for safety car probabilities, and responsive mobile-optimized layouts.

## 11. APPENDIX

**Source Code (Tableau Calculated Fields & Parameters)**

```
// Dynamic Top N Driver Ranking Filter
RANK(SUM([Points (Championship! Standings)]), 'desc') <= [Top N Parameter]

// Standardized DNF Reason Categorization
```

```
IF CONTAINS(LOWER([Status]), "engine") OR CONTAINS (LOWER ([Status]), "power unit") THEN "Engine /
Mechanical"
ELSEIF CONTAINS (LOWER([Status]), "gearbox") OR CONTAINS (LOWER([Status]), "transmission") THEN
"Gearbox"
ELSEIF CONTAINS (LOWER([Status]), "collision") OR CONTAINS (LOWER([Status]), "accident") THEN
"Collision / Incident"
ELSEIF CONTAINS (LOWER ([Status]), "brakes") OR CONTAINS (LOWER([Status]), "suspension") THEN
"Mechanical Failure"
ELSE "Other / Classified"
END

// Key Executive LOD Metrics
[Total DNFS]: IF [Status] != 'Finished' AND NOT (CONTAINS ([Status], '+')) THEN 1 ELSE END
[Circuits Showcased]: { FIXED : COUNTD ([Circuit Id]) }
[Competing Teams]: { FIXED : COUNTD ([Constructor Id]) }
[Seasons in Play]: { FIXED : COUNTD (YEAR([Race Date])) }
```

**Dataset & Project Links**

• **Dataset Source:** FIA Formula 1 Historical Telemetry Archive (2010-2026 seasons, 6,464 race records).
• **GitHub Repository:** https://github.com/Ashi0265/Formula1
• **Live Tableau Public Demo:** https://public.tableau.com/app/profile/rashi.khanvilkar/viz/ProjectF1_17935197172540/Story1
• **Demonstration Walkthrough:** 5-7 minute technical screen recording submitted to the SkillWallet platform.